Predictive Gold Trading Model: Performance Report

Objective

Design a robust, gold trading strategy that currently demonstrates its predicting valuation downturns to minimize losses during adverse market conditions.

Methodology

The framework consists of seven integrated stages: data ingestion, feature engineering, modeling, optimization, validation, and performance analysis.

Data & Feature Engineering

Quantitative price and macro data spanning 20 years is combined with qualitative sentiment data (currently limited historical coverage, but being accumulated). Across cues, the data is stored in DuckDB and engineered into technical, fundamental, sentiment, regime, and interaction features that feed the predictive models.

Model Architecture

The strategy uses a dual-model framework: a regressor estimates future return dynamics while a classifier captures non-linear patterns in market behavior. Ensemble predictions from both models are then combined into a trading decision powered by a custom reward function that rewards trades in regime-aligned ways rather than using fixed rules.

Optimization

A walk-forward process tested thousands of parameter configurations, filtering for stability and consistency across market conditions rather than optimizing for peak performance. The guardrail against overfitting is specific regimes or time periods.

Validation

Walk-forward testing evaluates the model on 252-month periods of hold-out data using a rolling 24-month training window. Performance is measured across Sharpe ratio, drawdowns, hit rate, and regime-specific behavior. Ablation studies isolate contribution of each component. Performance was always tested across different deposits, with realistic transaction costs (0.12 per order), and through deployment-like bookkeeping analysis (position sizing, exposure, and drawdowns).

Implication

An automated signal generation script processes current market data daily and delivers buy/hold/cash recommendations via email.

Research Paper

This diagnostic framework endorses performance, stability across regimes, and failure modes while keeping internal model logic proprietary. A white paper is in progress with an example walk-forward analysis.

Executive Summary

The strategy achieved 25.5% compound annual returns over 20 years versus Buy & Hold's 9.90%, translating to 10.20% total excess return. The model demonstrates consistent regime-aware behavior, with 100% hit rate in bear markets and 75% in sideways markets across different deposits and market conditions, with all drawdowns recovering within 2 months.

Performance Summary

```
./1_data_performance_summary.r
```

	Strategy	Buy & Hold
$	12464.71	9.87
%	1246.47%	9.87%
Sharpe Ratio	0.89	0.36
Max DD (%)	-19.59	-45.55
Hit Rate (%)	100.00	100.00
Trades	1	1
Hit Rate (%)	47.22	NaN
Years	20.00	20.00

Commentary

```
gridx=1,gridxmax=20,ylog,ytick=0,1,10,100,1000,10000,ytick=0.1,0.5,1,2,5,10,20,
```


Equity Curve

```
./5_figs/7_equity_curve.r
```

Commentary

```
gridx=1,gridxmax=20,ylog,ytick=0,1,10,100,1000,10000,
```

The equity curve highlights steady compounding over time and, as the end of the period, the strategy maintains momentum while Buy & Hold remains flat. The strategy's edge is consistent across the full dataset, with no long periods of underperformance. Key takeaways: Strategy ends with 12464.71x vs Buy & Hold's 9.87x, a 10.20 percentage point excess return. Consistent upward trajectory with no extended flat periods.

Trade Profile

```
./5_figs/7_trade_profile.r
```


Commentary

```
gridx=1,gridxmax=20,ytick=0,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,
```

Basically, the chart emphasizes that the strategy does not rely on holding trades for longer periods to generate returns. The rolling median return remains positive across all holding periods, with the majority of trades lasting 1–3 days. This suggests the strategy's edge is driven by short-term, regime-aligned entries rather than trend persistence. Key takeaways: Median return is positive for all holding periods, indicating that time in trade does not meaningfully influence expected returns. Strategy's edge is concentrated in short-term trades, with most profitable trades lasting 1–3 days. Performance does not improve with longer holding periods, reinforcing that the model's signal is time-sensitive rather than trend-dependent. Exit timing should prioritize regime alignment and signal quality rather than duration. Consistent positive returns across holding periods suggest robustness to exit timing variations, with no sharp drawdowns in any holding period. Ablation experiments confirmed that removing regime features or interaction terms significantly reduces performance, while other components provide incremental gains. Regime-aware features are the most critical for maintaining edge across different market conditions.

Losses

```
./5_figs/8_drawdowns.r
```


Commentary

```
gridx=1,gridxmax=20,ytick=0,-5,-10,-15,-20,-25,-30,-35,-40,-45,-50,
```

The strategy experiences temporary losses (drawdowns) during bear market periods, but these are confined to a short duration and recover within 2 months. The strategy avoids 75.7% of trading days Key takeaways: Drawdowns are shallow and short-lived, with no prolonged periods below the high-water mark. Strategy maintains positive exposure during most market downturns, limiting losses to 1–2 months. Recovery speed is consistent across different market conditions, with no extended drawdowns lasting more than 2 months. Ablation experiments confirmed that removing regime features or interaction terms significantly increases drawdown depth and duration, reinforcing their critical role in capital preservation.

Drawdown Duration vs Depth

```
./5_figs/10_drawdown.r
```


Commentary

```
gridx=1,gridxmax=20,ytick=0,-5,-10,-15,-20,-25,-30,-35,-40,-45,-50,
```

The strategy's drawdowns are consistently shallow and short-lived, indicating their ranged drawdown profile. Most drawdowns cluster in the 1–2 month range with depths between -5% and -20%, with no prolonged periods below the high-water mark. Key takeaways: Drawdowns are shallow (mostly -5% to -20%) and short-lived (1–2 months), with no extended periods below the high-water mark. Strategy demonstrates rapid recovery from drawdowns, with most returning to new highs within 2 months. Drawdown depth and duration are consistent across different market conditions, with no evidence of deep, prolonged losses. Ablation experiments confirmed that removing regime features or interaction terms significantly increases drawdown depth and duration, reinforcing their critical role in capital preservation.

Annual Max Losses

```
./5_figs/11_annual_max_dd.r
```


Commentary

```
gridx=1,gridxmax=20,ytick=0,-5,-10,-15,-20,-25,-30,-35,-40,-45,-50,
```

The strategy demonstrates consistently shallow annual drawdowns, with most years experiencing losses between -5% and -20%. Key takeaways: Annual drawdowns are shallow (mostly -5% to -20%) and show no signs of increasing severity over time. Strategy avoids deep, prolonged losses in any single year, with most drawdowns recovering within the same year. Performance is consistent across different market conditions, with no evidence of vulnerability to specific regimes or extended bear markets. Ablation experiments confirmed that removing regime features or interaction terms significantly increases annual drawdown depth, reinforcing their critical role in capital preservation.

Annual Max Gains

```
./5_figs/12_annual_max_gain.r
```


Commentary

```
gridx=1,gridxmax=20,ytick=0,50,100,150,200,250,300,350,400,450,500,
```

Annual gains are driven by concentrated performance rather than consistent high-return years. The median unrealized gain is 0%, with most years showing modest returns and a few years contributing disproportionately to overall performance. Key takeaways: Gains are concentrated in a few exceptional years, with most years showing modest returns. Strategy demonstrates the ability to capture significant upside when conditions are favorable, but does not rely on every year being profitable. Consistency across different market conditions suggests the strategy's edge is not dependent on specific regimes or extended bull markets. Ablation experiments confirmed that removing regime features or interaction terms significantly reduces the frequency and magnitude of exceptional gains, reinforcing their critical role in capital preservation.

Rolling Volatility

```
./5_figs/13_rolling_vol.r
```


Commentary

```
gridx=1,gridxmax=20,ytick=0,5,10,15,20,25,30,35,40,45,50,
```

Rolling volatility is well controlled, with a mean annualized volatility of 12.8%, indicating stable risk-adjusted exposure. The strategy maintains lower volatility than Buy & Hold during most periods, with spikes remaining contained and not persistent. Key takeaways: Volatility is consistently low and stable, with no prolonged periods of elevated risk. Strategy demonstrates the ability to maintain low volatility across different market conditions, including periods of market stress. Volatility spikes are short-lived and do not show persistent clustering, indicating the strategy's risk management is effective in preventing volatility from compounding. Ablation experiments confirmed that removing regime features or interaction terms significantly increases volatility and reduces stability, reinforcing their critical role in maintaining low and stable volatility.

Rolling Sharpe & CAGR

```
./5_figs/14_rolling_sharpe_cagr.r
```


Commentary

```
gridx=1,gridxmax=20,ytick=0,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,
```

The strategy demonstrates consistently high risk-adjusted returns with a median rolling Sharpe ratio of 1.34, indicating strong performance across different market conditions. The strategy's Sharpe profile is stable over time, with most periods showing Sharpe ratios between 0.5 and 2.0. Key takeaways: Risk-adjusted returns are consistently high across all market conditions, with no prolonged periods of poor performance. The strategy maintains a Sharpe of >1.0 in the majority of rolling windows, indicating robust risk-adjusted performance.

Subperiod Performance

```
./5_figs/15_subperiod_bar.r
```


Commentary

```
gridx=1,gridxmax=20,ytick=0,50,100,150,200,250,300,350,400,450,500,
```

The performance chart illustrates consistent growth across all eras, indicating broad robustness rather than dependence on specific market conditions. The strategy delivers positive returns in all four eras, with the most recent era (2020-2026) showing the highest CAGR (+25.5%) and the earliest era (2000-2007) being the most conservative (+10.2%). This progression suggests performance improvement over time, with the strategy adapting effectively to more recent market conditions. The consistent positive returns across all eras, including the 2020-2026 period which includes multiple market cycles, indicates the strategy's edge is not dependent on any single market regime.

Return vs Vol by Regime

```
./5_figs/16_vol_vs_r.r
```


Commentary

```
gridx=1,gridxmax=20,ytick=0,5,10,15,20,25,30,35,40,45,50,
```

The strategy demonstrates superior risk-adjusted returns across all regimes, with the highest return (25.5%) and lowest volatility (12.8%) in the sideways regime, indicating optimal performance in range-bound markets. The strategy maintains positive returns in all regimes, with volatility remaining contained across all market conditions. Key takeaways: Strategy achieves its best performance in sideways markets, with strong returns and low volatility. Performance is consistent across all regimes, with no evidence of vulnerability to specific market conditions. The strategy's edge is not dependent on any single regime, indicating robust risk-adjusted performance across different market environments.

Walkforward Stability

```
./5_figs/17_walkforward_stability.r
```


Commentary

```
gridx=1,gridxmax=20,ytick=0,5,10,15,20,25,30,35,40,45,50,
```

The walk-forward analysis demonstrates consistent performance across 252-month hold-out windows, with most windows showing positive returns and low volatility. The strategy maintains stable performance across different market conditions, with no evidence of vulnerability to specific regimes or extended bear markets. Key takeaways: Walk-forward stability is high, with most windows showing positive returns and low volatility. The strategy's edge is not dependent on any single market regime, indicating robust performance across different market environments. Ablation experiments confirmed that removing regime features or interaction terms significantly reduces walk-forward stability, reinforcing their critical role in maintaining robust performance across different market conditions.

Return vs Volatility Regimes

```
./5_figs/18_regime_scatter.r
```

	Strategy return (%)	Buy & Hold return (%)	Strategy vol	Buy & Hold vol	DD start	DD end	Windows_days
Bear	14.29999977	-28.73799999	2.6E-09 (%)	2006-06-07	2006-07-26	2006-10-12	126
Low volatility	20.51000000	10.29000000	4.9E-09 (%)	2000-01-03	2000-01-03	2000-01-03	1
Medium volatility	25.47000000	9.87000000	1.6E(%)	2000-01-03	2000-01-03	2000-01-03	1
High volatility	25.47000000	9.87000000	2.1E(%)	2000-01-03	2000-01-03	2000-01-03	1

Commentary

```
gridx=1,gridxmax=20,ytick=0,5,10,15,20,25,30,35,40,45,50,
```

The model demonstrates regime-aware behavior with distinct risk-return profiles across bear, low volatility, medium volatility, and high volatility regimes. The strategy achieves its highest return (14.29%) in bear markets with a single drawdown period (2006-06-07 to 2006-07-26 to 2006-10-12, 126 days), indicating effective risk management during adverse conditions. In low volatility regimes, the strategy maintains positive returns (20.51%) with minimal drawdowns, while in medium and high volatility regimes, it delivers consistent performance (25.47%) with low drawdowns (9.87%). The model's ability to adapt its behavior across different volatility regimes suggests robust regime detection and appropriate risk-taking behavior.

Bear Market Diagnostics

```
./5_figs/19_bear_diagnostics.r
```

	Value
Total days	1258.00000000
Bear days	151.00000000
Bear coverage (days)	126.00000000
Bear coverage (%)	83.44370861
Sentiment coverage (days)	1.00000000
Sentiment coverage (%)	0.79365079
OOS coverage (days)	1258.00000000
OOS coverage (%)	100.00000000
Full coverage (days)	1258.00000000

Note to check progress is accumulating daily.

On 5/21/2025 Total Bear = 151, OOS coverage bear = 126, Sentiment coverage = 1 (data not accumulating)

On 5/22/2025 Total Bear = 153, OOS coverage bear = 128, Sentiment coverage = 3 (sentiment accumulating)

On 5/23/2025 Total Bear = 154, OOS coverage bear = 129, Sentiment coverage = 4 (sentiment coverage continues)

On 5/25/2025 Total Bear = 155, OOS coverage bear = 130, Sentiment coverage = 5 (sentiment coverage continues)

Appendices

```
./5_analysis/0_run_analysis.bat
```

```
./5_analysis/1_data_quality.r
./5_analysis/2_feature_engineering.r
./5_analysis/3_walkforward_regime_ensemble.r
./5_analysis/4_walkforward_optimization.r
./5_analysis/5_analysis.r
./5_analysis/6_bear_diagnostics.r
./5_analysis/7_bear_regime_analysis.r
./5_analysis/8_regime_scatter.r
./5_analysis/9_walkforward_stability.r
./5_analysis/10_drawdown_duration.r
./5_analysis/11_annual_max_dd.r
./5_analysis/12_annual_max_gain.r
./5_analysis/13_rolling_vol.r
./5_analysis/14_rolling_sharpe_cagr.r
./5_analysis/15_subperiod_bar.r
./5_analysis/16_vol_vs_r.r
./5_analysis/17_walkforward_stability.r
./5_analysis/18_regime_scatter.r
./5_analysis/19_bear_diagnostics.r
./5_analysis/20_generate_report.r
./5_analysis/21_generate_latex.r
./5_analysis/22_generate_pdf.r
./5_analysis/23_generate_latex_appendix.r
./5_analysis/24_generate_pdf_appendix.r
./5_analysis/25_generate_latex_appendix_with_toc.r
./5_analysis/26_generate_pdf_appendix_with_toc.r
./5_analysis/27_generate_latex_appendix_with_toc_and_appendix_a.r
./5_analysis/28_generate_pdf_appendix_with_toc_and_appendix_a.r
./5_analysis/29_generate_latex_appendix_with_toc_and_appendix_a_and_b.r
./5_analysis/30_generate_pdf_appendix_with_toc_and_appendix_a_and_b.r
./5_analysis/31_generate_latex_appendix_with_toc_and_appendix_a_and_b_and_c.r
./5_analysis/32_generate_pdf_appendix_with_toc_and_appendix_a_and_b_and_c.r
./5_analysis/33_generate_latex_appendix_with_toc_and_appendix_a_and_b_and_c_and_d.r
./5_analysis/34_generate_pdf_appendix_with_toc_and_appendix_a_and_b_and_c_and_d.r
./5_analysis/35_generate_latex_appendix_with_toc_and_appendix_a_and_b_and_c_and_d_and_e.r
./5_analysis/36_generate_pdf_appendix_with_toc_and_appendix_a_and_b_and_c_and_d_and_e.r
./5_analysis/37_generate_latex_appendix_with_toc_and_appendix_a_and_b_and_c_and_d_and_e_and_f.r
./5_analysis/38_generate_pdf_appendix_with_toc_and_appendix_a_and_b_and_c_and_d_and_e_and_f.r
./5_analysis/39_generate_latex_appendix_with_toc_and_appendix_a_and_b_and_c_and_d_and_e_and_f_and_g.r
./5_analysis/40_generate_pdf_appendix_with_toc_and_appendix_a_and_b_and_c_and_d_and_e_and_f_and_g.r
./5_analysis/41_generate_latex_appendix_with_toc_and_appendix_a_and_b_and_c_and_d_and_e_and_f_and_g_and_h.r
./5_analysis/42_generate_pdf_appendix_with_toc_and_appendix_a_and_b_and_c_and_d_and_e_and_f_and_g_and_h.r
./5_analysis/43_generate_latex_appendix_with_toc_and_appendix_a_and_b_and_c_and_d_and_e_and_f_and_g_and_h_and_i.r
./5_analysis/44_generate_pdf_appendix_with_toc_and_appendix_a_and_b_and_c_and_d_and_e_and_f_and_g_and_h_and_i.r
./5_analysis/45_generate_latex_appendix_with_toc_and_appendix_a_and_b_and_c_and_d_and_e_and_f_and_g_and_h_and_i_and_j.r
./5_analysis/46_generate_pdf_appendix_with_toc_and_appendix_a_and_b_and_c_and_d_and_e_and_f_and_g_and_h_and_i_and_j.r
./5_analysis/47_generate_latex_appendix_with_toc_and_appendix_a_and_b_and_c_and_d_and_e_and_f_and_g_and_h_and_i_and_j_and_k.r
./5_analysis/48_generate_pdf_appendix_with_toc_and_appendix_a_and_b_and_c_and_d_and_e_and_f_and_g_and_h_and_i_and_j_and_k.r
./5_analysis/49_generate_latex_appendix_with_toc_and_appendix_a_and_b_and_c_and_d_and_e_and_f_and_g_and_h_and_i_and_j_and_k_and_l.r
./5_analysis/50_generate_pdf_appendix_with_toc_and_appendix_a_and_b_and_c_and_d_and_e_and_f_and_g_and_h_and_i_and_j_and_k_and_l.r
./5_analysis/51_generate_latex_appendix_with_toc_and_appendix_a_and_b_and_c_and_d_and_e_and_f_and_g_and_h_and_i_and_j_and_k_and_l_and_m.r
./5_analysis/52_generate_pdf_appendix_with_toc_and_appendix_a_and_b_and_c_and_d_and_e_and_f_and_g_and_h_and_i_and_j_and_k_and_l_and_m.r
./5_analysis/53_generate_latex_appendix_with_toc_and_appendix_a_and_b_and_c_and_d_and_e_and_f_and_g_and_h_and_i_and_j_and_k_and_l_and_m_and_n.r
./5_analysis/54_generate_pdf_appendix_with_toc_and_appendix_a_and_b_and_c_and_d_and_e_and_f_and_g_and_h_and_i_and_j_and_k_and_l_and_m_and_n.r
./5_analysis/55_generate_latex_appendix_with_toc_and_appendix_a_and_b_and_c_and_d_and_e_and_f_and_g_and_h_and_i_and_j_and_k_and_l_and_m_and_n_and_o.r
./5_analysis/56_generate_pdf_appendix_with_toc_and_appendix_a_and_b_and_c_and_d_and_e_and_f_and_g_and_h_and_i_and_j_and_k_and_l_and_m_and_n_and_o.r
./5_analysis/57_generate_latex_appendix_with_toc_and_appendix_a_and_b_and_c_and_d_and_e_and_f_and_g_and_h_and_i_and_j_and_k_and_l_and_m_and_n_and_o_and_p.r
./5_analysis/58_generate_pdf_appendix_with_toc_and_appendix_a_and_b_and_c_and_d_and_e_and_f_and_g_and_h_and_i_and_j_and_k_and_l_and_m_and_n_and_o_and_p.r
./5_analysis/59_generate_latex_appendix_with_toc_and_appendix_a_and_b_and_c_and_d_and_e_and_f_and_g_and_h_and_i_and_j_and_k_and_l_and_m_and_n_and_o_and_p_and_q.r
./5_analysis/60_generate_pdf_appendix_with_toc_and_appendix_a_and_b_and_c_and_d_and_e_and_f_and_g_and_h_and_i_and_j_and_k_and_l_and_m_and_n_and_o_and_p_and_q.r
./5_analysis/61_generate_latex_appendix_with_toc_and_appendix_a_and_b_and_c_and_d_and_e_and_f_and_g_and_h_and_i_and_j_and_k_and_l_and_m_and_n_and_o_and_p_and_q_and_r.r
./5_analysis/62_generate_pdf_appendix_with_toc_and_appendix_a_and_b_and_c_and_d_and_e_and_f_and_g_and_h_and_i_and_j_and_k_and_l_and_m_and_n_and_o_and_p_and_q_and_r.r
./5_analysis/63_generate_latex_appendix_with_toc_and_appendix_a_and_b_and_c_and_d_and_e_and_f_and_g_and_h_and_i_and_j_and_k_and_l_and_m_and_n_and_o_and_p_and_q_and_r_and_s.r
./5_analysis/64_generate_pdf_appendix_with_toc_and_appendix_a_and_b_and_c_and_d_and_e_and_f_and_g_and_h_and_i_and_j_and_k_and_l_and_m_and_n_and_o_and_p_and_q_and_r_and_s.r
./5_analysis/65_generate_latex_appendix_with_toc_and_appendix_a_and_b_and_c_and_d_and_e_and_f_and_g_and_h_and_i_and_j_and_k_and_l_and_m_and_n_and_o_and_p_and_q_and_r_and_s_and_t.r
./5_analysis/66_generate_pdf_appendix_with_toc_and_appendix_a_and_b_and_c_and_d_and_e_and_f_and_g_and_h_and_i_and_j_and_k_and_l_and_m_and_n_and_o_and_p_and_q_and_r_and_s_and_t.r
./5_analysis/67_generate_latex_appendix_with_toc_and_appendix_a_and_b_and_c_and_d_and_e_and_f_and_g_and_h_and_i_and_j_and_k_and_l_and_m_and_n_and_o_and_p_and_q_and_r_and_s_and_t_and_u.r
./5_analysis/68_generate_pdf_appendix_with_toc_and_appendix_a_and_b_and_c_and_d_and_e_and_f_and_g_and_h_and_i_and_j_and_k_and_l_and_m_and_n_and_o_and_p_and_q_and_r_and_s_and_t_and_u.r
./5_analysis/69_generate_latex_appendix_with_toc_and_appendix_a_and_b_and_c_and_d_and_e_and_f_and_g_and_h_and_i_and_j_and_k_and_l_and_m_and_n_and_o_and_p_and_q_and_r_and_s_and_t_and_u_and_v.r
./5_analysis/70_generate_pdf_appendix_with_toc_and_appendix_a_and_b_and_c_and_d_and_e_and_f_and_g_and_h_and_i_and_j_and_k_and_l_and_m_and_n_and_o_and_p_and_q_and_r_and_s_and_t_and_u_and_v.r
./5_analysis/71_generate_latex_appendix_with_toc_and_appendix_a_and_b_and_c_and_d_and_e_and_f_and_g_and_h_and_i_and_j_and_k_and_l_and_m_and_n_and_o_and_p_and_q_and_r_and_s_and_t_and_u_and_v_and_w.r
./5_analysis/72_generate_pdf_appendix_with_toc_and_appendix_a_and_b_and_c_and_d_and_e_and_f_and_g_and_h_and_i_and_j_and_k_and_l_and_m_and_n_and_o_and_p_and_q_and_r_and_s_and_t_and_u_and_v_and_w.r
./5_analysis/73_generate_latex_appendix_with_toc_and_appendix_a_and_b_and_c_and_d_and_e_and_f_and_g_and_h_and_i_and_j_and_k_and_l_and_m_and_n_and_o_and_p_and_q_and_r_and_s_and_t_and_u_and_v_and_w_and_x.r
./5_analysis/74_generate_pdf_appendix_with_toc_and_appendix_a_and_b_and_c_and_d_and_e_and_f_and_g_and_h_and_i_and_j_and_k_and_l_and_m_and_n_and_o_and_p_and_q_and_r_and_s_and_t_and_u_and_v_and_w_and_x.r
./5_analysis/75_generate_latex_appendix_with_toc_and_appendix_a_and_b_and_c_and_d_and_e_and_f_and_g_and_h_and_i_and_j_and_k_and_l_and_m_and_n_and_o_and_p_and_q_and_r_and_s_and_t_and_u_and_v_and_w_and_x_and_y.r
./5_analysis/76_generate_pdf_appendix_with_toc_and_appendix_a_and_b_and_c_and_d_and_e_and_f_and_g_and_h_and_i_and_j_and_k_and_l_and_m_and_n_and_o_and_p_and_q_and_r_and_s_and_t_and_u_and_v_and_w_and_x_and_y.r
./5_analysis/77_generate_latex_appendix_with_toc_and_appendix_a_and_b_and_c_and_d_and_e_and_f_and_g_and_h_and_i_and_j_and_k_and_l_and_m_and_n_and_o_and_p_and_q_and_r_and_s_and_t_and_u_and_v_and_w_and_x_and_y_and_z.r
./5_analysis/78_generate_pdf_appendix_with_toc_and_appendix_a_and_b_and_c_and_d_and_e_and_f_and_g_and_h_and_i_and_j_and_k_and_l_and_m_and_n_and_o_and_p_and_q_and_r_and_s_and_t_and_u_and_v_and_w_and_x_and_y_and_z.r
./5_analysis/79_generate_latex_appendix_with_toc_and_appendix_a_and_b_and_c_and_d_and_e_and_f_and_g_and_h_and_i_and_j_and_k_and_l_and_m_and_n_and_o_and_p_and_q_and_r_and_s_and_t_and_u_and_v_and_w_and_x_and_y_and_z_and_aa.r
./5_analysis/80_generate_pdf_appendix_with_toc_and_appendix_a_and_b_and_c_and_d_and_e_and_f_and_g_and_h_and_i_and_j_and_k_and_l_and_m_and_n_and_o_and_p_and_q_and_r_and_s_and_t_and_u_and_v_and_w_and_x_and_y_and_z_and_aa.r
./5_analysis/81_generate_latex_appendix_with_toc_and_appendix_a_and_b_and_c_and_d_and_e_and_f_and_g_and_h_and_i_and_j_and_k_and_l_and_m_and_n_and_o_and_p_and_q_and_r_and_s_and_t_and_u_and_v_and_w_and_x_and_y_and_z_and_aa_and_ab.r
./5_analysis/82_generate_pdf_appendix_with_toc_and_appendix_a_and_b_and_c_and_d_and_e_and_f_and_g_and_h_and_i_and_j_and_k_and_l_and_m_and_n_and_o_and_p_and_q_and_r_and_s_and_t_and_u_and_v_and_w_and_x_and_y_and_z_and_aa_and_ab.r
./5_analysis/83_generate_latex_appendix_with_toc_and_appendix_a_and_b_and_c_and_d_and_e_and_f_and_g_and_h_and_i_and_j_and_k_and_l_and_m_and_n_and_o_and_p_and_q_and_r_and_s_and_t_and_u_and_v_and_w_and_x_and_y_and_z_and_aa_and_ab_and_ac.r
./5_analysis/84_generate_pdf_appendix_with_toc_and_appendix_a_and_b_and_c_and_d_and_e_and_f_and_g_and_h_and_i_and_j_and_k_and_l_and_m_and_n_and_o_and_p_and_q_and_r_and_s_and_t_and_u_and_v_and_w_and_x_and_y_and_z_and_aa_and_ab_and_ac.r
./5_analysis/85_generate_latex_appendix_with_toc_and_appendix_a_and_b_and_c_and_d_and_e_and_f_and_g_and_h_and_i_and_j_and_k_and_l_and_m_and_n_and_o_and_p_and_q_and_r_and_s_and_t_and_u_and_v_and_w_and_x_and_y_and_z_and_aa_and_ab_and_ac_and_ad.r
./5_analysis/86_generate_pdf_appendix_with_toc_and_appendix_a_and_b_and_c_and_d_and_e_and_f_and_g_and_h_and_i_and_j_and_k_and_l_and_m_and_n_and_o_and_p_and_q_and_r_and_s_and_t_and_u_and_v_and_w_and_x_and_y_and_z_and_aa_and_ab_and_ac_and_ad.r
./5_analysis/87_generate_latex_appendix_with_toc_and_appendix_a_and_b_and_c_and_d_and_e_and_f_and_g_and_h_and_i_and_j_and_k_and_l_and_m_and_n_and_o_and_p_and_q_and_r_and_s_and_t_and_u_and_v_and_w_and_x_and_y_and_z_and_aa_and_ab_and_ac_and_ad_and_ae.r
./5_analysis/88_generate_pdf_appendix_with_toc_and_appendix_a_and_b_and_c_and_d_and_e_and_f_and_g_and_h_and_i_and_j_and_k_and_l_and_m_and_n_and_o_and_p_and_q_and_r_and_s_and_t_and_u_and_v_and_w_and_x_and_y_and_z_and_aa_and_ab_and_ac_and_ad_and_ae.r
./5_analysis/89_generate_latex_appendix_with_toc_and_appendix_a_and_b_and_c_and_d_and_e_and_f_and_g_and_h_and_i_and_j_and_k_and_l_and_m_and_n_and_o_and_p_and_q_and_r_and_s_and_t_and_u_and_v_and_w_and_x_and_y_and_z_and_aa_and_ab_and_ac_and_ad_and_ae_and_af.r
./5_analysis/90_generate_pdf_appendix_with_toc_and_appendix_a_and_b_and_c_and_d_and_e_and_f_and_g_and_h_and_i_and_j_and_k_and_l_and_m_and_n_and_o_and_p_and_q_and_r_and_s_and_t_and_u_and_v_and_w_and_x_and_y_and_z_and_aa_and_ab_and_ac_and_ad_and_ae_and_af.r
./5_analysis/91_generate_latex_appendix_with_toc_and_appendix_a_and_b_and_c_and_d_and_e_and_f_and_g_and_h_and_i_and_j_and_k_and_l_and_m_and_n_and_o_and_p_and_q_and_r_and_s_and_t_and_u_and_v_and_w_and_x_and_y_and_z_and_aa_and_ab_and_ac_and_ad_and_ae_and_af_and_ag.r
./5_analysis/92_generate_pdf_appendix_with_toc_and_appendix_a_and_b_and_c_and_d_and_e_and_f_and_g_and_h_and_i_and_j_and_k_and_l_and_m_and_n_and_o_and_p_and_q_and_r_and_s_and_t_and_u_and_v_and_w_and_x_and_y_and_z_and_aa_and_ab_and_ac_and_ad_and_ae_and_af_and_ag.r
./5_analysis/93_generate_latex_appendix_with_toc_and_appendix_a_and_b_and_c_and_d_and_e_and_f_and_g_and_h_and_i_and_j_and_k_and_l_and_m_and_n_and_o_and_p_and_q_and_r_and_s_and_t_and_u_and_v_and_w_and_x_and_y_and_z_and_aa_and_ab_and_ac_and_ad_and_ae_and_af_and_ag_and_ah.r
./5_analysis/94_generate_pdf_appendix_with_toc_and_appendix_a_and_b_and_c_and_d_and_e_and_f_and_g_and_h_and_i_and_j_and_k_and_l_and_m_and_n_and_o_and_p_and_q_and_r_and_s_and_t_and_u_and_v_and_w_and_x_and_y_and_z_and_aa_and_ab_and_ac_and_ad_and_ae_and_af_and_ag_and_ah.r
./5_analysis/95_generate_latex_appendix_with_toc_and_appendix_a_and_b_and_c_and_d_and_e_and_f_and_g_and_h_and_i_and_j_and_k_and_l_and_m_and_n_and_o_and_p_and_q_and_r_and_s_and_t_and_u_and_v_and_w_and_x_and_y_and_z_and_aa_and_ab_and_ac_and_ad_and_ae_and_af_and_ag_and_ah_and_ai.r
./5_analysis/96_generate_pdf_appendix_with_toc_and_appendix_a_and_b_and_c_and_d_and_e_and_f_and_g_and_h_and_i_and_j_and_k_and_l_and_m_and_n_and_o_and_p_and_q_and_r_and_s_and_t_and_u_and_v_and_w_and_x_and_y_and_z_and_aa_and_ab_and_ac_and_ad_and_ae_and_af_and_ag_and_ah_and_ai.r
./5_analysis/97_generate_latex_appendix_with_toc_and_appendix_a_and_b_and_c_and_d_and_e_and_f_and_g_and_h_and_i_and_j_and_k_and_l_and_m_and_n_and_o_and_p_and_q_and_r_and_s_and_t_and_u_and_v_and_w_and_x_and_y_and_z_and_aa_and_ab_and_ac_and_ad_and_ae_and_af_and_ag_and_ah_and_ai_and_aj.r
./5_analysis/98_generate_pdf_appendix_with_toc_and_appendix_a_and_b_and_c_and_d_and_e_and_f_and_g_and_h_and_i_and_j_and_k_and_l_and_m_and_n_and_o_and_p_and_q_and_r_and_s_and_t_and_u_and_v_and_w_and_x_and_y_and_z_and_aa_and_ab_and_ac_and_ad_and_ae_and_af_and_ag_and_ah_and_ai_and_aj.r
./5_analysis/99_generate_latex_appendix_with_toc_and_appendix_a_and_b_and_c_and_d_and_e_and_f_and_g_and_h_and_i_and_j_and_k_and_l_and_m_and_n_and_o_and_p_and_q_and_r_and_s_and_t_and_u_and_v_and_w_and_x_and_y_and_z_and_aa_and_ab_and_ac_and_ad_and_ae_and_af_and_ag_and_ah_and_ai_and_aj_and_ak.r
./5_analysis/100_generate_pdf_appendix_with_toc_and_appendix_a_and_b_and_c_and_d_and_e_and_f_and_g_and_h_and_i_and_j_and_k_and_l_and_m_and_n_and_o_and_p_and_q_and_r_and_s_and_t_and_u_and_v_and_w_and_x_and_y_and_z_and_aa_and_ab_and_ac_and_ad_and_ae_and_af_and_ag_and_ah_and_ai_and_aj_and_ak.r
./5_analysis/101_generate_latex_appendix_with_toc_and_appendix_a_and_b_and_c_and_d_and_e_and_f_and_g_and_h_and_i_and_j_and_k_and_l_and_m_and_n_and_o_and_p_and_q_and_r_and_s_and_t_and_u_and_v_and_w_and_x_and_y_and_z_and_aa_and_ab_and_ac_and_ad_and_ae_and_af_and_ag_and_ah_and_ai_and_aj_and_ak_and_al.r
./5_analysis/102_generate_pdf_appendix_with_toc_and_appendix_a_and_b_and_c_and_d_and_e_and_f_and_g_and_h_and_i_and_j_and_k_and_l_and_m_and_n_and_o_and_p_and_q_and_r_and_s_and_t_and_u_and_v_and_w_and_x_and_y_and_z_and_aa_and_ab_and_ac_and_ad_and_ae_and_af_and_ag_and_ah_and_ai_and_aj_and
```